

Aman Anwar

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EDUCATION

Virginia Tech
B.S. in Computer Science

Blacksburg, VA
May 2028

TECHNICAL SKILLS

Java, JavaScript, TypeScript, Python, SQL, Docker, Flask, Node.js, PyTorch, Scikit-Learn, Pandas, React.js, REST APIs, Express, MongoDB, PostgreSQL, Windows Command Line, Linux Command Line, Bash Scripting, Google Cloud Platform

EXPERIENCE

IDOpt Lab @ Virginia Tech
Undergraduate Researcher

Blacksburg, VA
Jan. 2026 – Present

- Contributing to the NASA RAM project by engineering high-quality documentation for the lab's research initiatives and packaged existing software as a reusable Python package.
- Drew insights from the Woods and Poole dataset and visualized year-wise county-wise demographic trends on an interactive map.
- Trained a machine learning model on origin-destination pairs from Bureau of Transportation Statistics data to predict fares between any two points in the network.

VTHacks
Technical Lead

Blacksburg, VA
Jan. 2026 – Present

- Lead a 6-member development team to build a real-time hackathon management system supporting 650+ active participants.

IDPRO @ Virginia Tech
Research Project Lead

Blacksburg, VA
Aug. 2025 – Present

- Leading a team of 5 in cutting-edge spatial audio research, collaborating with an industry-leading sound engineer to explore automation of Spatial Audio (ambisonics).
- Developed an end-to-end audio processing pipeline integrating OpenAI Whisper, Facebook DEMUCS, and LLM fine-tuning on Apple MLX for intelligent sound separation and scene adaptation deployed on AWS Bedrock.
- Authored a successful proposal securing a \$1,000 ICAT SEAD Grant to support ongoing research.

Lumiq
Software Engineer Intern

Remote
May 2025 – Aug. 2025

- Built an end-to-end churn prediction pipeline using Python and Scikit-learn, achieving a 95% F1-score in forecasting customer churn.
- Performed extensive exploratory data analysis, feature engineering, and SMOTE-based resampling on 100K+ customer records to correct class imbalance and enhance minority recall by 40%.
- Automated retraining and reporting workflows with Streamlit, improving early churn detection accuracy by 25% and enabling actionable retention insights.

IDEEAS LAB @ Virginia Tech
Undergraduate Researcher

Blacksburg, VA
Feb. 2025 – Aug. 2025

- Engineering a novel spatial audio synthesis system using Gemini 2.5 AI models, enabling more immersive and individualized 3D audio experiences.
- Developed a Python spatialization library, `spatialAudiopy`, utilizing object-oriented design principles for intuitive spatial audio development.
- Increased research productivity by 2x using custom developed libraries over traditional methods.

PROJECTS

- Dime** | *React, Flask, Snowflake, Photon iMessageKit* | HackViolet 2026 Jan. 2026
- Built a credit card reward optimization platform with a browser extension, web dashboard, and iMessage agent that automatically tracks rewards and nudges users to use the optimal card at checkout.
 - Integrated KnotAPI's TransactionLink for secure transaction data fetching and Snowflake Cortex AI for transaction categorization, powering real-time spend analytics and reward optimization at scale.
 - Developed Photon iMessage agent for conversational transaction ingestion via receipt scanning using Gemini, enabling users to chat with their financial data entirely through text.
- PiggyBank** | *Knot API, Photon iMessageKit, Flask, React* | 3 Awards @ HackPrinceton 2025 Nov. 2025
- Built a financial agent that learns users' financial habits to predict users' next transactions to nudge them via iMessage to help develop better financial habits.
 - Integrated Knot's Production TransactionLink with Photon's iMessage Kit to deliver real transactional insights and nudges using a custom prediction model deployed on DigitalOcean, powered by vector search in Snowflake.
 - Awarded 2nd Place by Knot, Best use of DigitalOcean by MLH, and Honorable mention by Photon Framework.
- Harvey the Helper** | *Python, Gemini, Quartz* | VTHacks 13 Sept. 2025
- Built an AI voice agent using Gemini 2.5 Flash that takes control of a user's computer based on verbal requests, enabling hands-free operation for elderly and disabled users.
 - Engineered a three-stage pipeline using Gemini for conversational interface, action extraction from dialogue transcripts, and real-time screen analysis to execute keyboard and mouse actions via Python Quartz library.
 - Implemented grid-based coordinate system for accurate UI element targeting, solving visual misinterpretation challenges in screenshot-based interaction.
- Questionable** | *React, TypeScript, Flask, Perplexity* | Perplexity Hackathon Aug. 2025
- Built a full-stack interactive news platform using React, TypeScript, and Flask, enhancing engagement with AI-powered content widgets that transform passive reading into active learning.
 - Integrated Perplexity's Sonar Deep Research API to generate topic-based question-headline pairs and Sonar Pro for full story rendering with interactive widgets including sliders and misconception reveals.
 - Reduced API latency by 60% through MongoDB caching with 24-hour TTL indexing and designed an animation-rich UI using Framer Motion and Tailwind CSS.
- Atlas** | *React, Flask, Google Cloud Platform* | 2nd Place @ Marriott Codefest 2025 Feb. 2025
- Built an infinite-canvas trip planner using React Flow, Flask, and MongoDB Atlas, enabling visual, non-linear travel organization with automatic state saving and real-time sync.
 - Integrated AI-driven personalization through RAG and vector search, plus Google Places API for accurate place feeds and Nano Banana model for intuitive room requests.
 - Strengthened platform security via JWT authentication, bcrypt hashing, and hard-to-guess upload URLs, and delivered analytics dashboards helping hotels refine marketing and guest experience.
- Fit Finder** | *JavaScript, Perplexity, MongoDB, Python* | Winner @ HooHacks25 Mar. 2025
- Built a visual outfit similarity search engine using vector embeddings from OpenCLIP (ChromaDB) and vector search from scalable MongoDB Atlas, enabling personalized apparel discovery in user uploaded closet pictures.
 - Integrated Google Gemini to generate outfit metadata and segments, boosting vector search accuracy by 2x.
 - Increased visual product search accuracy by 50% by integrating Perplexity Sonar with Gemini-generated image descriptions.
- Mumbl** | *JavaScript, Node.js, Express, Gemini* | HackViolet 2025 Feb. 2025
- Developed a web-based speech coaching platform that provides advanced presentation prompts and real-time analysis of speech delivery using Google Gemini for audio processing.
 - Built full-stack application with Express backend API for grading and feedback generation, enabling users to improve presentation skills through AI-powered insights.
 - Implemented interactive chatbot for post-presentation Q&A, allowing users to ask detailed questions about their performance analysis.
- Questionable** | *React, TypeScript, Flask* | Perplexity Hackathon May 2025
- Built a full-stack interactive news platform using React, TypeScript, and Flask, enhancing engagement with AI-powered content widgets.
 - Integrated Perplexity's Sonar models into a scalable Flask backend with cached API endpoints for real-time generated insights.
 - Reduced API latency by 60% through MongoDB caching and designed an animation-rich UI using Framer Motion and Tailwind CSS.

AWARDS & RESEARCH GRANTS

1st Place Overall | 2025 CMDA Fall Data Competition

Academy of Data Science, Virginia Tech

Blacksburg, VA

Nov. 2025

2nd Place | Knot Track @ HackPrinceton

KnotAPI

New York, NY

Nov. 2025

Best Use of Digital Ocean @ HackPrinceton

MLH

New York, NY

Nov. 2025

Honorable Mention @ HackPrinceton

Photon Framework

New York, NY

Nov. 2025

2nd Place Overall @ Marriott Codefest

Marriott International

Bethesda, MD

Oct. 2025

Roger & Debbie West ICAT SEAD Grant

Virginia Tech Institute for Creativity, Arts, and Technology

Blacksburg, VA

Oct. 2025

Best Use of MongoDB Atlas @ MLH | HooHacks25

University of Virginia

Charlottesville, VA

Mar. 2025